

Jotham Teshome

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EDUCATION

Michigan State University

Master of Science, Computer Science

January 2023 - August 2024

GPA: 3.85

- Relevant Coursework: Natural Language Processing, Computer Vision, Pattern Recognition, Deep Learning, Algorithmic Graph Theory, Distributed Systems

Michigan State University

Bachelor of Science, Computer Science, Minor in Business

September 2019 - December 2022

GPA: 3.69

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Algorithm Engineering, Web Application Development, Database Systems

WORK EXPERIENCE

Stefanini Group - AI Software Engineer

April 2025 - Present

- Prototyped a **local LLM**-based API replacement for **OpenAI** in the **MDR** pipeline, deploying on a remote machine on-premises to meet client data privacy requirements
- Extended a **.NET** cybersecurity MDR platform that ingests **SIEM** alerts, normalizes them via **Elastic Common Schema (ECS)** over **RabbitMQ**, and automatically creates tickets in a connected **ITSM** system across multiple enterprise clients
- Built a **neural network** false-positive classifier for SIEM incidents, leveraging **OpenAI**-generated JSON reports for feature extraction and reducing analyst workload on false alerts
- Contributed features to a **Vue.js** analyst platform including **role-based access control**, per-client ITSM/SIEM health-check dashboards, and a full frontend refactor to align with company design standards
- Containerized services with **Docker**, stored images in **Azure Container Registry (ACR)**, and deployed via **Kubernetes** across a multi-tenant, multi-client architecture
- Created a globally distributed **Python** tool to automate **SAP** document creation, cutting generation time by **70%**, and automated the pipeline with **FastAPI**, **RabbitMQ**, and **Azure Blob Storage**

Versive - Machine Learning Engineer

June 2024 - December 2024

- Built **CNN** models in **PyTorch** to detect and classify tumors in MRI scans, achieving strong accuracy on validation data
- Experimented with **YOLOv8 segmentation** on a small dataset to evaluate region-based detection, improving performance by **10%**
- Explored **transfer learning** with pretrained **ImageNet** weights to optimize training speed, reducing training time by **30%**

Versive - Software Engineer

January 2023 - July 2023

- Developed a secure hospital app with **AWS Amplify** and **RDS**, improving MRI access times and reducing verification time by **20%**
- Built **Flask** APIs connecting a **MySQL** backend with a **React** frontend, and ensured HIPAA compliance through encryption, **RBAC**, and **MFA**
- Leveraged **Docker** containers and **Git** for deployment and version control, improving development cycle efficiency by **15%**

MSU Federal Credit Union - Software Engineer Intern

September 2022 - December 2022

- Designed a peer-to-peer transfer system in **Flutter/Dart** with QR code and NFC support, increasing user transaction speed by **25%**
- Integrated **Google Places API** to send personalized deal notifications based on shopping patterns, boosting app engagement by **18%**
- Contributed to app improvements using **Flutter**, **Dart**, and **SQL**, enhancing overall app stability and reducing crash rates by **12%**

CERTIFICATIONS

AWS Certified AI Practitioner - Amazon Web Services (AWS)

June 2026

PROJECTS

Identifying and Removing Toxic Comments, Python, TensorFlow

- Built an **RNN** model in **TensorFlow** to detect toxicity in online comments, designing the model architecture and training pipeline
- Generated subword embeddings using **FastText** to better detect potential variations of toxic words in comments
- Achieved a word-level classification accuracy of **91%** using trigram embeddings in a multi-appearance word model
- Developed an automated system to censor toxic words, improving the efficiency of real-time content moderation

Watch Together, React, TypeScript, Node.js, Express, Socket.IO, AWS

- Built a real-time collaborative video platform syncing **YouTube** playback across multiple clients using **WebSockets** via **Socket.IO**
- Implemented latency tracking and drift correction on the **Node.js/Express** backend to keep all users frame-accurate during playback
- Managed distributed room state server-side, handling join/rejoin sync and propagating play, pause, and seek events in real time
- Deployed on **AWS** via **EC2**, **CloudFront**, **S3**, and **Route53** with an automated **GitHub Actions** CI/CD pipeline

SKILLS

AI/ML:	PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, YOLO, OpenCV, FastText, OpenAI API
Programming:	Python, C/C++, C#, Java, JavaScript, TypeScript, HTML, CSS, Dart
Web & App Dev:	React, Vue.js, Next.js, Node.js, Express, Flask, FastAPI, Flutter, .NET
Databases:	SQL, MySQL, PostgreSQL, SQL Server, Redis, DynamoDB, MongoDB
Cloud & DevOps:	AWS, Azure, GCP, Docker, Kubernetes, GitHub Actions, RabbitMQ, ACR
Tools & Platforms:	Linux, Unix, Bash, Git, Postman, SAP, WebSockets